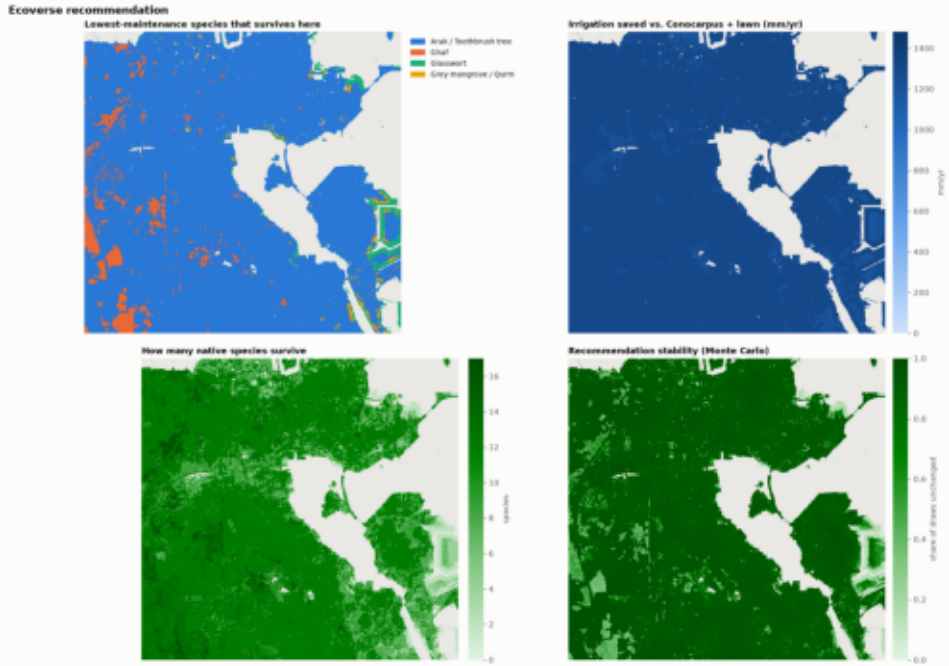
Ecoverse 813

Plant what the ground already wants.

Satellite environmental fingerprints that pick the lowest-maintenance native plants able to survive each patch of urban Bahrain.



Theme 05 · Sustainable Urban Planning & Smart Cities

Country: Kingdom of Bahrain

Team: Sara Abutaleb, Enam Abutaleb

THE PROBLEM

Bahrain greens its cities by fighting its own soil

- Reference evapotranspiration ≈ 1,850 mm/yr against ≈ 75 mm of rain: most urban plantings survive only with irrigation (Fanack Water; FAO AQUASTAT).
- Soils are saline and calcareous: irrigated soils 4–12 dS/m, abandoned farmland up to 60 dS/m, shallow saline groundwater on the coast (FAO AQUASTAT).
- The default urban tree, Conocarpus, was banned for new planting in 2022 because its roots break pipes, drains and walls (GDN Online).
- Species are still chosen from generic lists, not from the conditions of the actual site — so planners overpay in water, maintenance and failed plantings.
- Who is affected: municipalities and their landscaping budgets, the water authority, and residents of the hottest, least-green neighbourhoods.

1,108 ha

of irrigated greenery detected in our area of interest

150 km²

of land analysed at 20 m resolution

21 species

native or Gulf-adapted, scored against 5 common urban plantings

BUSINESS USE CASE

Who uses Ecoverse and what decision it changes

User

Municipal landscaping planners (Ministry of Municipalities Affairs & Agriculture, the four municipalities), real-estate developers and landscape contractors.

Decision

Which species — canopy, shrub, groundcover — to plant or replant on a given site, and which existing irrigated plantings to convert first.

Today

Standard species lists, nursery availability and habit; failures are masked by extra irrigation and maintenance.

What Ecoverse delivers

A district map of the lowest-maintenance guild that survives each 20 m patch; a one-page site card per project; the water it saves; a field-probe route that targets the uncertain sites.

Why it scales

Runs on free Sentinel-2/Landsat now, Satellite 813 hyperspectral later; the ground kit costs ≈ USD 100; one analyst can cover a municipality. Revenue: per-district subscription or per-project site report.

ORIGINALITY

What already exists — and what is new here

Native-plant guides

Gulf botanic and municipal lists (e.g. Qatar, UAE) say WHICH species are adapted, but not WHERE each one fits; they are static PDFs.

Tree-canopy & heat tools

Urban forestry tools say WHERE to add trees (heat, equity) but assume a temperate species palette and irrigation.

Agricultural salinity mapping

Biosaline programmes map salinity to match crops on farms; they target yield, not low-input urban landscapes.

- Ecoverse joins the three: a per-pixel fingerprint → a species GUILD that survives with minimal water → the water it saves.
- Hyperspectral over Bahrain: NASA EMIT measures saline gypsum soil directly, and shows what Sentinel-2 misses.
- Uncertainty-driven field loop: the map tells two students where one probe reading changes the most decisions, then updates itself.

DATA

Open data only — and where hyperspectral earns its place

- Sentinel-2 L2A (ESA Copernicus): 42 clear summer scenes, 2025-06-01/2025-08-31, 20 m; plus a winter composite.
- Landsat 8/9 Collection 2 surface temperature (USGS): summer heat.
- Copernicus DEM GLO-30: elevation → shallow saline groundwater risk.
- ESA WorldCover 2021: land-cover context and an independent mangrove reference.
- NASA EMIT hyperspectral (285 bands, 60 m) over the same Bahrain AOI: NASA EMIT EMIT_L2A_RFL_002_20230617T111139 (2023-06-17).
- 26-species trait table from FAO, ICBA, peer-reviewed salinity trials and Gulf floras (sources per row).

Hyperspectral signal recoverable from Sentinel-2 (spatial-block cross-validation)

GYPSUM_1750: CV R^2 = 0.43

CLAY_2200: CV R^2 = 0.27

CARBONATE_2340: CV R^2 = 0.39

REP: CV R^2 = 0.65

CANOPY_WATER_1200: CV R^2 = 0.85

EMIT gypsum (1750 nm) used directly in the Bahrain salinity layer.

Sentinel-2 separates the 6 hyperspectral soil classes with balanced accuracy 0.58 (chance 0.17).

WORKFLOW

Approach

1 · Open data

S2 summer/winter, Landsat LST, DEM, WorldCover, clipped to the AOI

2 · Hyperspectral (EMIT)

Gypsum, clay, carbonate, plant stress over Bahrain; what S2 can recover

3 · Fingerprint

heat · salinity (ECe) · saline groundwater · moisture · survival evidence · urban density

4 · Survival model

26 species × factor response curves → survival score

5 · Guild optimiser

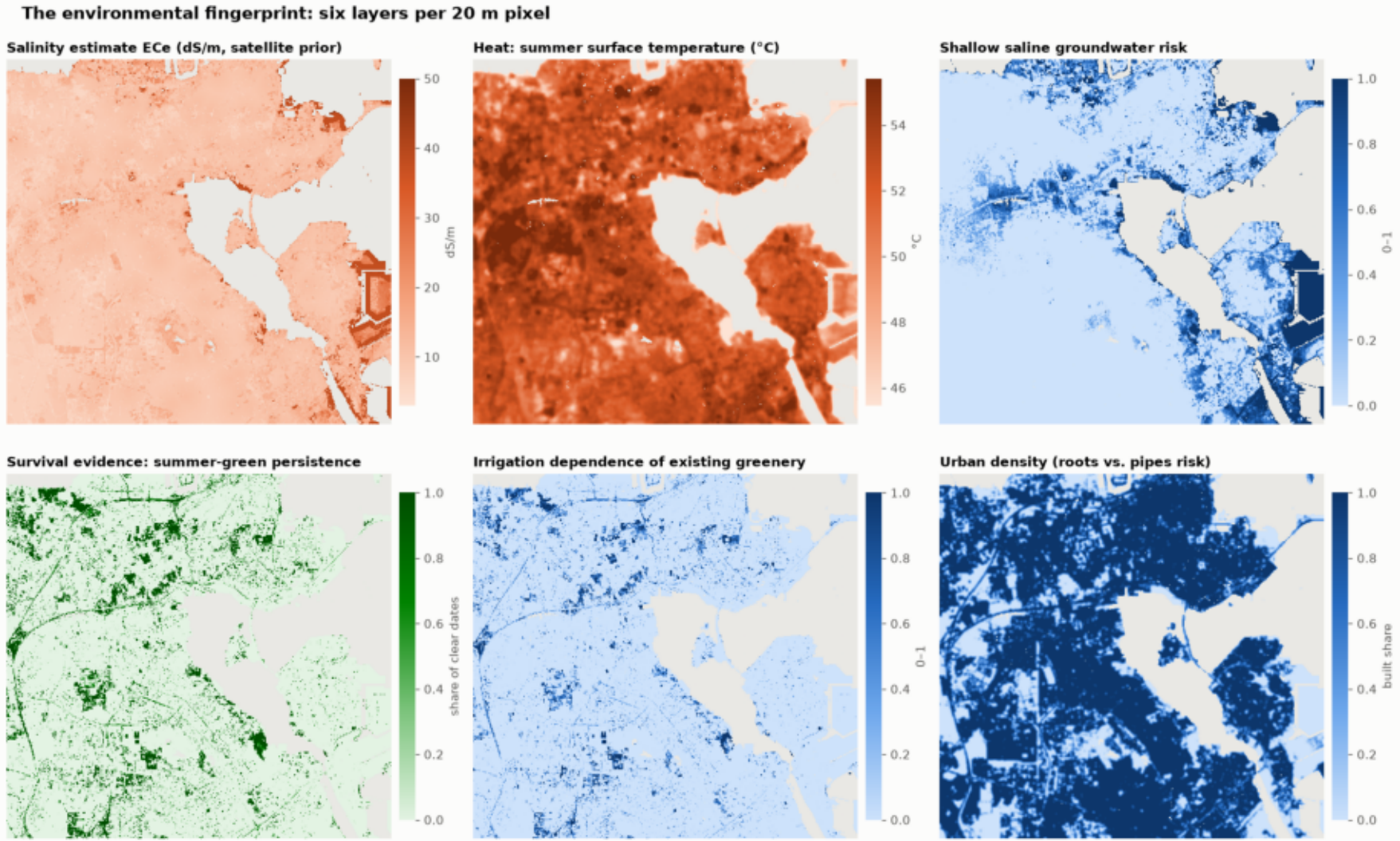
lowest water + upkeep, shade where hottest

6 · Robustness & probes

Monte Carlo stability → probe route → map update

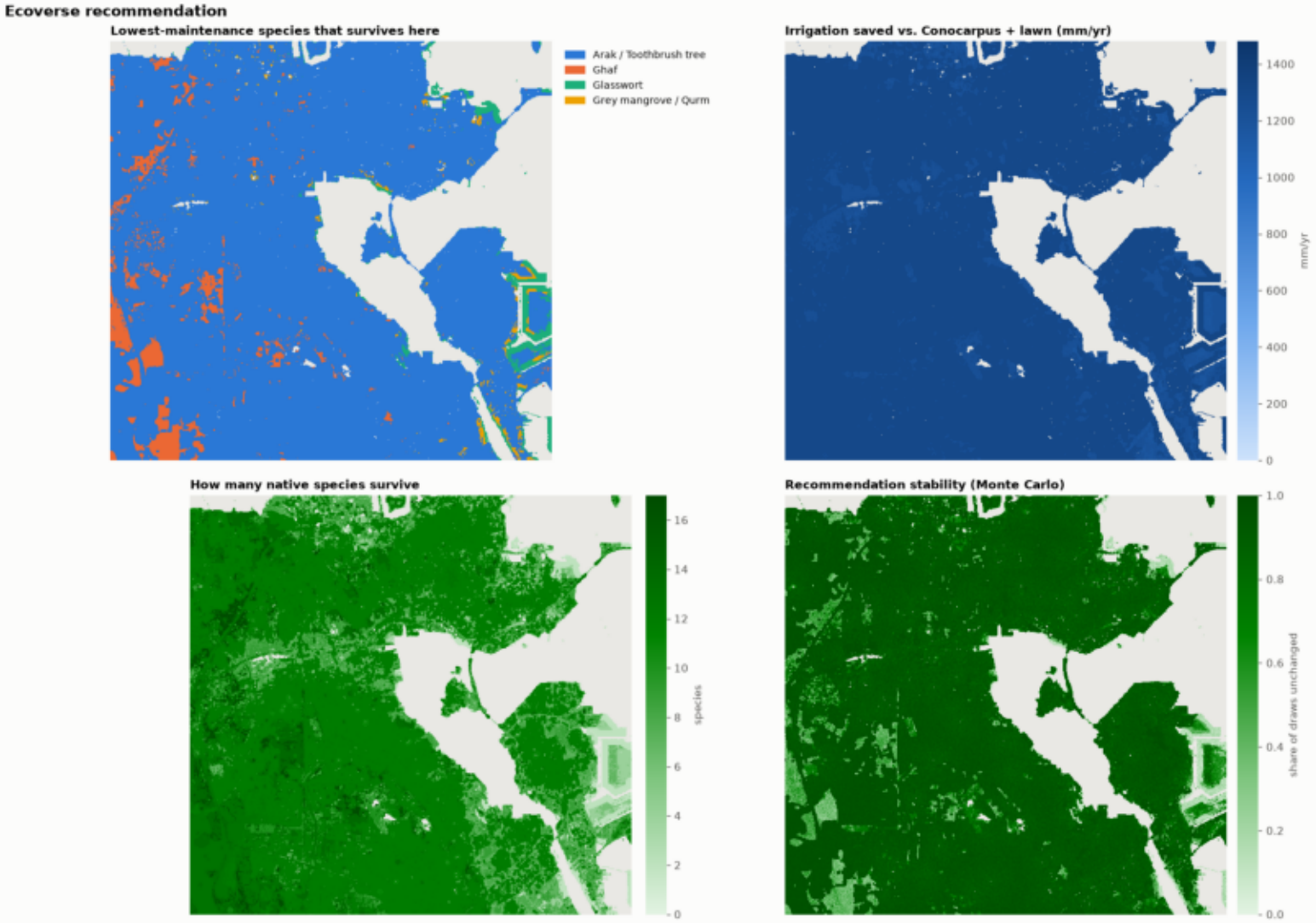
OUTPUT 1

The environmental fingerprint of every 20 m patch



OUTPUT 2

What to plant where — and how sure we are



Most-recommended natives

Arak / Toothbrush tree: 97% of land

Ghaf: 77% of land

Rimth: 44% of land

Harm / Qatar bean caper: 34% of land

Sea lavender: 14% of land

100%

of land has at least one native that survives without routine irrigation

88%

of land keeps the same pick in ≥80% of Monte Carlo draws

OUTPUT 3

The site card a planner actually reads

Hot dense urban block

Summer surface	60.4 °C
Salinity (prior)	11 dS/m (5–23)
Saline groundwater	0.00
Urban density	1.00
Greenery now	none / sparse

Recommended guild

Canopy	Ghaf
Shrub	Arak / Toothbrush tree
Ground	Harm / Qatar bean caper

Water need 185 mm/yr

Saves 1295 mm/yr (= 12950 m³ per ha)

Conocarpus survival score: 0.12

Saline open ground

Summer surface	47.7 °C
Salinity (prior)	39 dS/m (27–57)
Saline groundwater	1.00
Urban density	0.00
Greenery now	none / sparse

Recommended guild

Canopy	none survives
Shrub	none survives
Ground	Glasswort

Water need 370 mm/yr

Saves 1110 mm/yr (= 11100 m³ per ha)

Conocarpus survival score: 0.00

Irrigated urban greenery

Summer surface	51.3 °C
Salinity (prior)	7 dS/m (2–21)
Saline groundwater	0.00
Urban density	1.00
Greenery now	irrigated

Recommended guild

Canopy	Ghaf
Shrub	Arak / Toothbrush tree
Ground	Rimth

Water need 185 mm/yr

Saves 1295 mm/yr (= 12950 m³ per ha)

Conocarpus survival score: 0.15

Coastal / intertidal edge

Summer surface	44.6 °C
Salinity (prior)	49 dS/m (22–109)
Saline groundwater	1.00
Urban density	0.00
Greenery now	tidal

Recommended guild

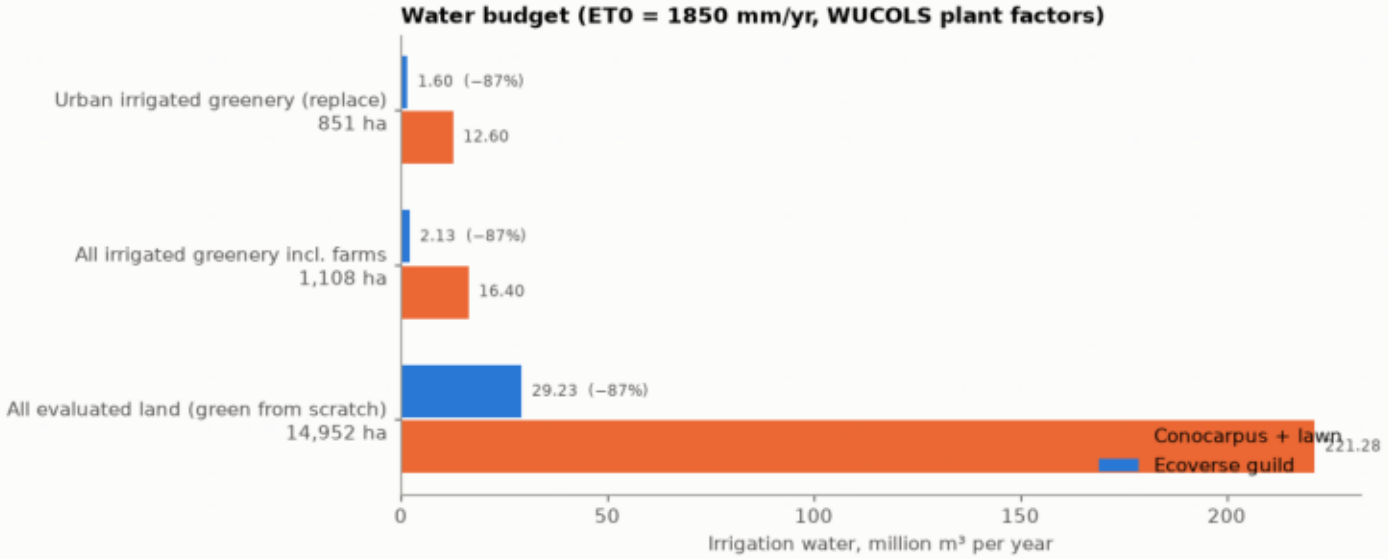
Canopy	none survives
Shrub	none survives
Ground	none survives

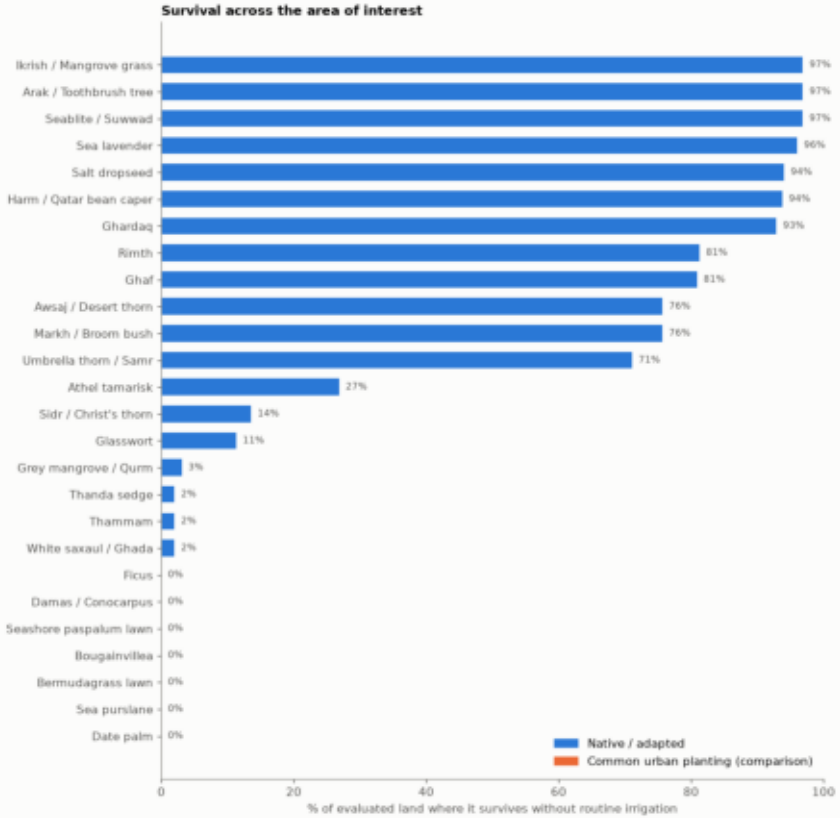
No species survives without irrigation: leave as natural sabkha or use engineered shade instead of planting.

Conocarpus survival score: 0.00

OUTPUT 4

Water saved vs. Conocarpus + lawn





11.00 M m³/yr

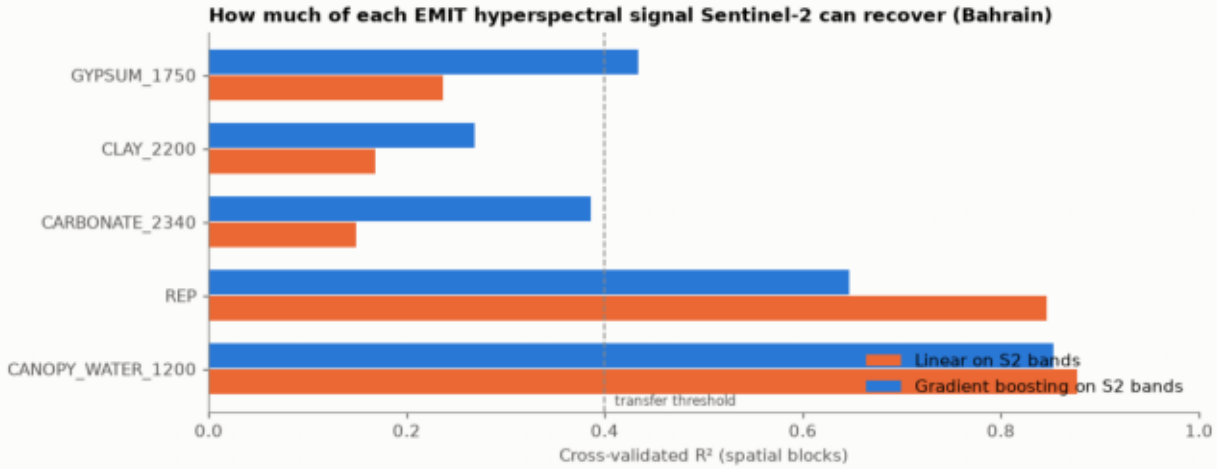
saved by converting the 851 ha of urban irrigated greenery in this AOI (87% less water)

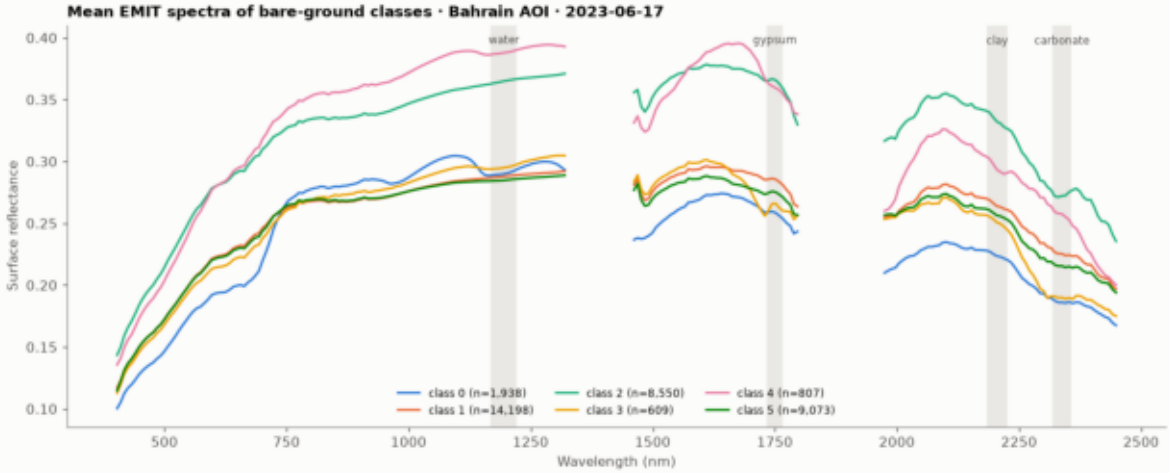
Mean saving 1285 mm/yr per planted m² (90% Monte Carlo interval 1249–1432).

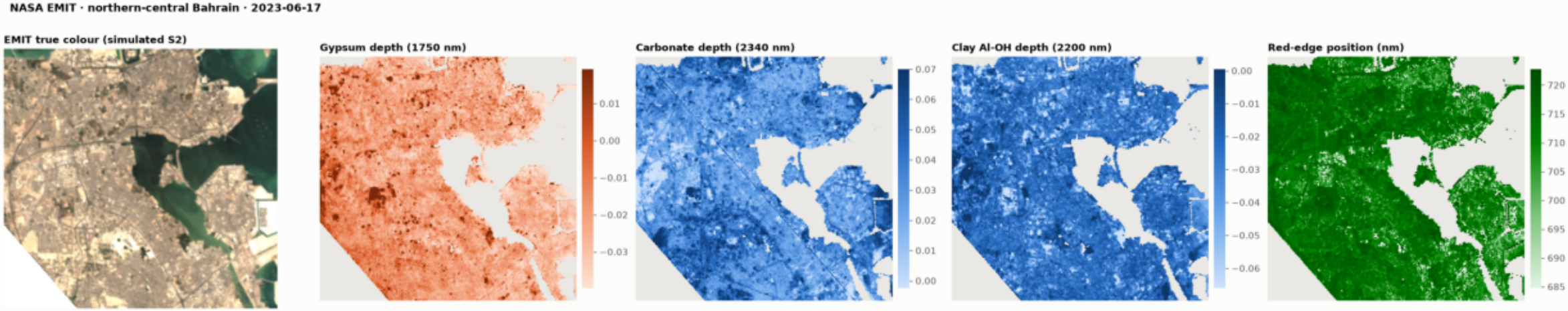
OUTPUT 5

Hyperspectral: what Sentinel-2 cannot see

Same Bahrain AOI, NASA EMIT EMIT_L2A_RFL_002_20230617T111139 (2023-06-17). The gypsum layer in the salinity map is measured by EMIT; bars show how much of each hyperspectral signal free Sentinel-2 can recover (the rest needs a hyperspectral sensor).

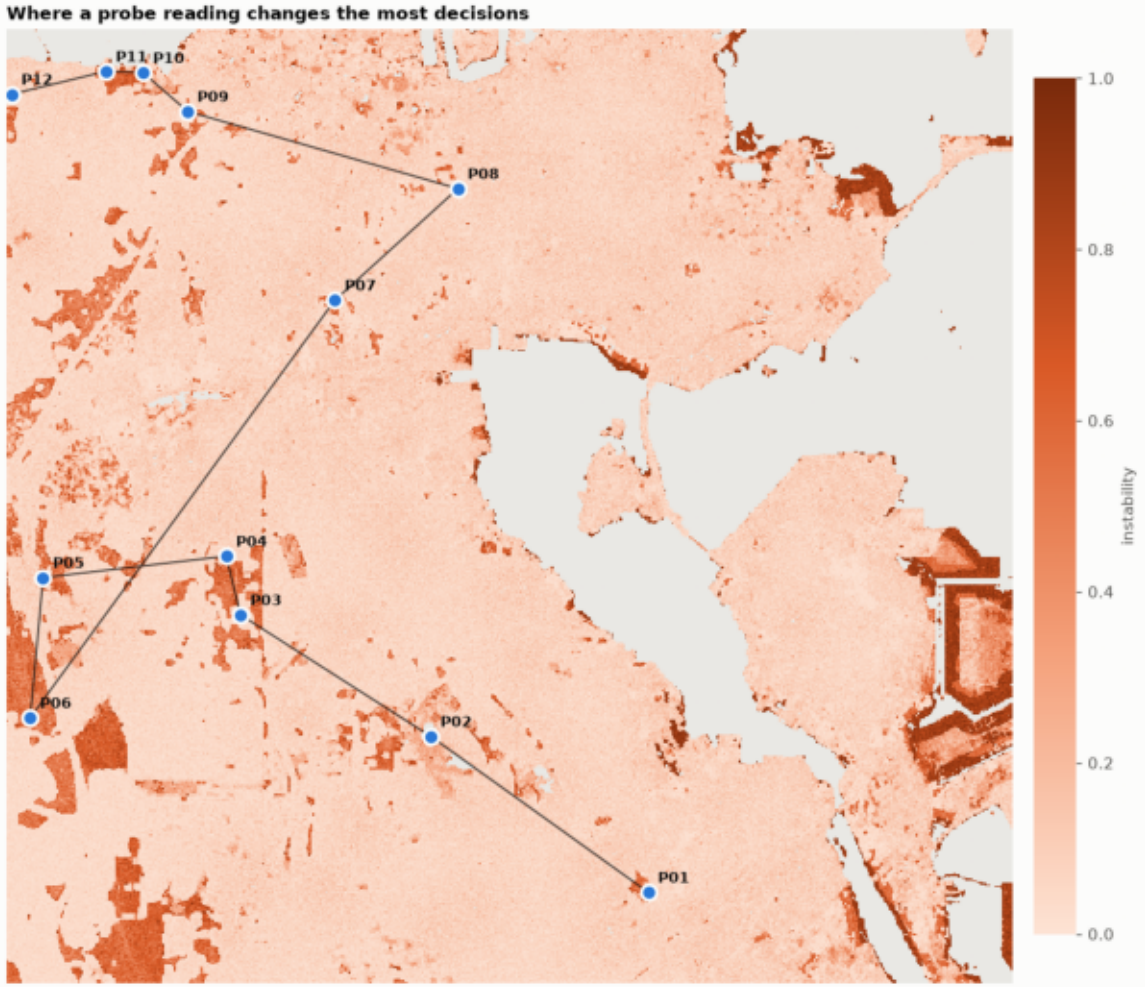


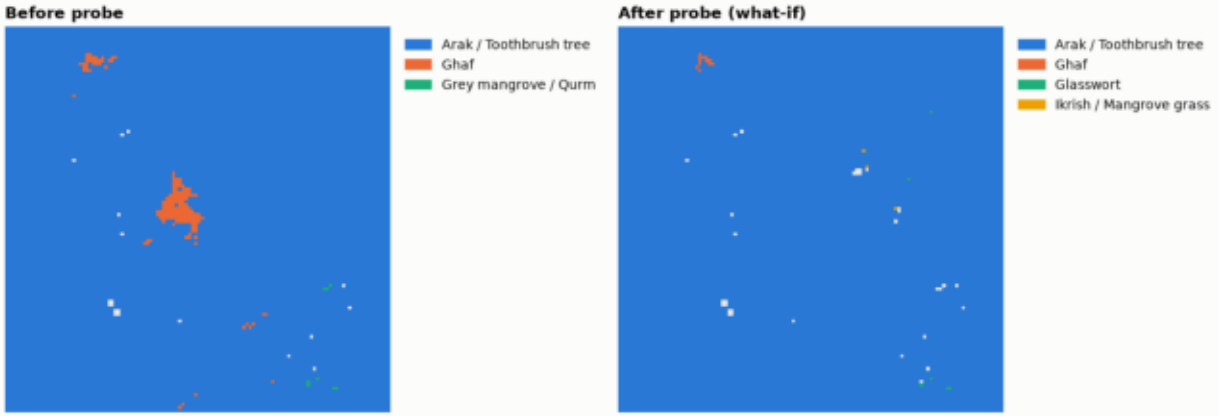




OUTPUT 6

Closing the loop: satellites tell students where to measure





What-if with a HYPOTHETICAL reading: the map updates around the probe

- 12 sites chosen where a reading flips the most decisions, ordered into one route.
- Kit ≈ USD 100: EC 1:5 probe, pH probe, capacitive moisture, NPK sensor, IR thermometer, phone GPS.
- Readings fuse into the map (residual kriging) and shrink uncertainty locally.

HONESTY SLIDE

Validation and limitations

Checks we can run today

- WorldCover mangrove pixels where grey mangrove survives (%): 100.00
- Inland (>1 km from sea) pixels where grey mangrove is recommended (%): 0.00
- Spearman rho, summer NDVI vs LST (expect < 0): -0.07 (weak at 20 m; LST is 100 m Landsat resampled)
- Conocarpus survives without routine irrigation in dense urban pixels (%): 0.00 (follows from its low drought rating: a consistency check, not independent evidence)
- Pixels with ≥1 surviving native species (%): 99.99
- EMIT coverage of the AOI grid (%): 95.68
- EMIT vs Sentinel-2 red reflectance, Pearson r (co-registration check): 0.80
- EMIT vs Sentinel-2 NIR reflectance, Pearson r (co-registration check): 0.89
- Median Monte Carlo stability: 0.93

Limitations

- Salinity is a satellite prior anchored to FAO ranges, not a measurement — probes calibrate it.
- Several species tolerances are expert estimates; a Gulf horticulturist must review the table.
- pH and nutrients are invisible from orbit; a regional prior is used.
- 20 m pixels mix roads, roofs and verges.
- Hyperspectral data is one EMIT date (June 2023) at 60 m; Sentinel-2 is 2025 at 20 m.
- Water savings assume established plants and WUCOLS plant factors.

WHY IT MATTERS

Impact

87%

modelled irrigation saving where natives replace high-water plantings

21

native or Gulf-adapted species brought into urban planting decisions

11.0 M m³

modelled water saved per year by converting this AOI's urban irrigated greenery

- Water security: less desalinated / groundwater spent on ornamentals.
- Lower maintenance budgets: fewer replacements, less pruning and fertiliser.
- Cooler streets: shade-giving natives are prioritised where surfaces are hottest.
- Biodiversity and heritage: ghaf, sidr, arak and halophytes return to the city.
- Transferable to every Gulf city with the same open data and a ≈ USD 100 kit.

ROADMAP

Next steps for incubation

Satellite 813

Repeat, higher-resolution hyperspectral coverage of Bahrain to track soil salinity and plant stress over time, beyond a single EMIT date.

Field campaign

50 probe sites + lab ECe on 10 samples to calibrate the salinity prior; survival audit of existing plantings.

Expert review

Gulf horticulturists and nurseries validate the trait table and add availability and cost per species.

Pilot

One municipality, one district: plant two demonstration verges (Ecoverse guild vs. standard) and measure water and survival for 12 months.

Product

Web map + site-card generator on gIQ; upload probe readings from a phone; export planting schedules for contractors.